

2026 INNOVATION CUP FOR AI AGENTS

Craft solutions
with AI and win!

Merck KGaA, Darmstadt,
Germany



CHALLENGE 1

Identify conserved and tissue-specific mechanisms of pathological wound healing across chronic fibrotic diseases, and prioritize intervention hypotheses that may reduce fibrosis and shift tissue responses towards repair and regeneration.



CHALLENGE 2

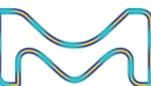
Predict the hydrophobicity interaction chromatography (HIC) retention time of 81 VHH antibody fragments using sequence information only

This CSV file contains 81 VHH amino acid sequences. Your goal is to predict the HIC retention time (in minutes) for each sequence.

The submitted paper must contain a table with two columns: sequence-id and predicted-HIC. Submissions will be scored using Pearson and Spearman correlation between your predictions and the true experimental HIC values.

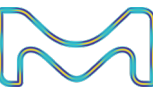
Use only the provided sequences and publicly available resources.

Submissions must be reproducible.



CHALLENGE 2

ID	VHH sequence
1	EVQLLESGGGVVQPGGSLRLSCAASGRTLSSYVMGWFRQAPGKEREFVSAISSSGDRYYADSVKGRFTISRDNAKNTVYLQMNSLRAEDTAVYYCAAALAPSGTLVVVSPLGYTYWGQGLTVTVSS
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CHALLENGE 2

41 DVQLQESGGGVVQPGGSLRLSCAASGRTFSSYVMGWFRQAPGKEREFSVAISWGGDNTYYADSVKGRFTISRDNANTVYLQMNSLRAEDTAVYYCAAALAPSGTTVVVSPLGYDNWGQGTQVTVSS

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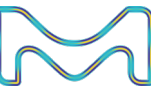
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CHALLENGE 3

Design a Novel Linker Chemistry with in-silico proof points

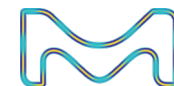
Background

Antibody-drug conjugates (ADCs) are a class of targeted therapeutics that fuse the precision of monoclonal antibodies (mAbs) with the high potency of payloads. ADCs offer several strategic advantages for the drug industry: 1. enable targeted delivery to diseased tissues to improve the therapeutic index and reduce off-target toxicity; 2. allow the use of highly potent payloads that would be too toxic systemically; 3. provide modular platforms for biomarker-driven patient selection, supporting precision medicine. An ADC's performance hinges on three integrated components: the antibody for tumor targeting, the payload for bioactivity, and the linker that connects the payload to the antibody. Although the linker itself does not exhibit the bioactivity or antigen selectivity, it largely determines clinical performance of ADC. The linker governs plasma stability, pharmacokinetics, and the conditional release of the payload at the tumor site. Cleavable linkers respond to tumor microenvironment cues (acidic pH, proteases, or reducing environments) to release payload after internalization. Non-cleavable linkers rely on antibody degradation, producing different pharmacology and potentially altered bystander effects. Payload diversification beyond traditional cytotoxins is expanding the ADC field's scope and demanding compatibility with broader conjugation chemistries and trigger mechanisms. This diversification aims to broaden therapeutic windows, overcome resistance, and enable combination strategies.

Problem statement

Many ADCs suffer from suboptimal drug properties (e.g., instability, aggregation, poor solubility) and pharmacology (narrow therapeutic windows, systemic toxicity, and insufficient tumor payload delivery) largely due to premature payload release or insufficient release at the tumor site. Existing linkers often lack tunability to accommodate diverse antibody-payload combinations and cannot adapt to heterogeneous tumor environments. There is a critical need for next-generation linkers that deliver enhanced in vivo stability, precise and tunable payload release, and broader compatibility with non-traditional payloads.

Key objectives for linker design include balancing circulation stability with efficient tumor- or cell-type-specific release while preserving antibody binding and Fc functions. Goals also encompass developing tunable cleavage triggers, improving aqueous solubility, and enabling robust conjugation chemistries across diverse payload classes (including oligonucleotides and immunomodulators). Achieving scalable, cost-efficient manufacturing with consistent product quality will require integrated linker technologies that enable rapid, reproducible production and broader therapeutic applicability across heterogeneous tumor environments.

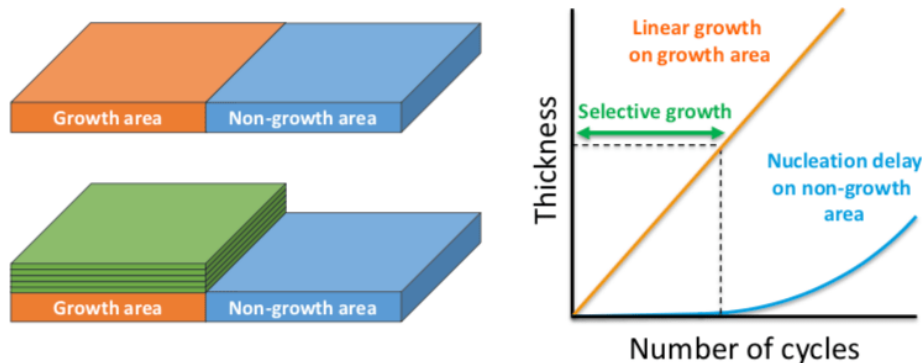


CHALLENGE 4

In-silico AI co-scientist for selective atomic layer growth schemes

Area selective atomic layer deposition (AS-ALD) offers precise, conformal thin film growth only on targeted regions of a semiconductor wafer, reducing the need for complex patterning and etching steps. This improves material usage efficiency, enhances device performance by minimizing defects, and enables finer feature control critical for advanced semiconductor manufacturing.

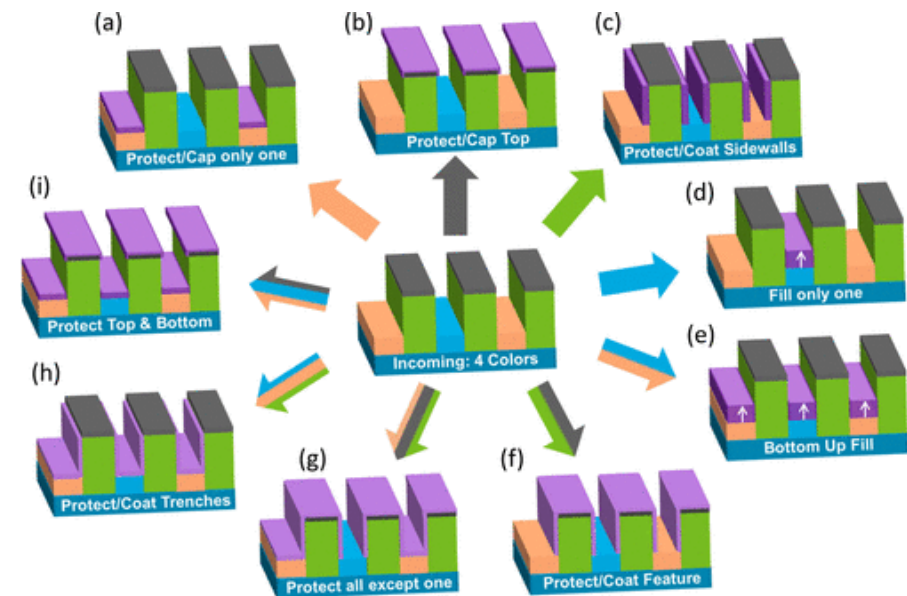
AS-ALD Process



<https://semiengineering.com/what-happened-to-selective-deposition/>

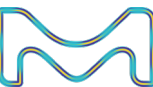
Area Selective Atomic Layer Deposition, process of selectively depositing on one area (Growth Surface or GS) and not depositing on another area (Non-Growth Surface or NGS). This can be done in many ways- intrinsically, creation of passivation layers, controlled reaction pathways, etc. The ideal outcome is to passivate the NGS such that it inhibits the growth and the GS has minimal defects. This can be done through tuning the surface chemistry and inhibitor/precursor chemistry.

AS-ALD Applications



Chem. Mater. 2020, 32, 4920–4953

Schematic representation of the "selective deposition toolbox" illustrating the types of structures that ASD could provide given a generic incoming structure with four unspecified materials exposed. By selectively depositing on one or more materials (colors) a variety of structures can be realized. For example, images a–d show ASD on only one color, and images e and f show ASD on two out of four colors. Likewise, images g–i show ASD on three colors. Other color combinations can also be imagined.



CHALLENGE 4

In-silico AI co-scientist for selective atomic layer growth schemes

**Growth
surface
(GS)**

**Non-growth
surface
(NGS)**



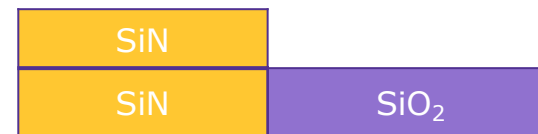
1. Pre-clean to differentiate surfaces



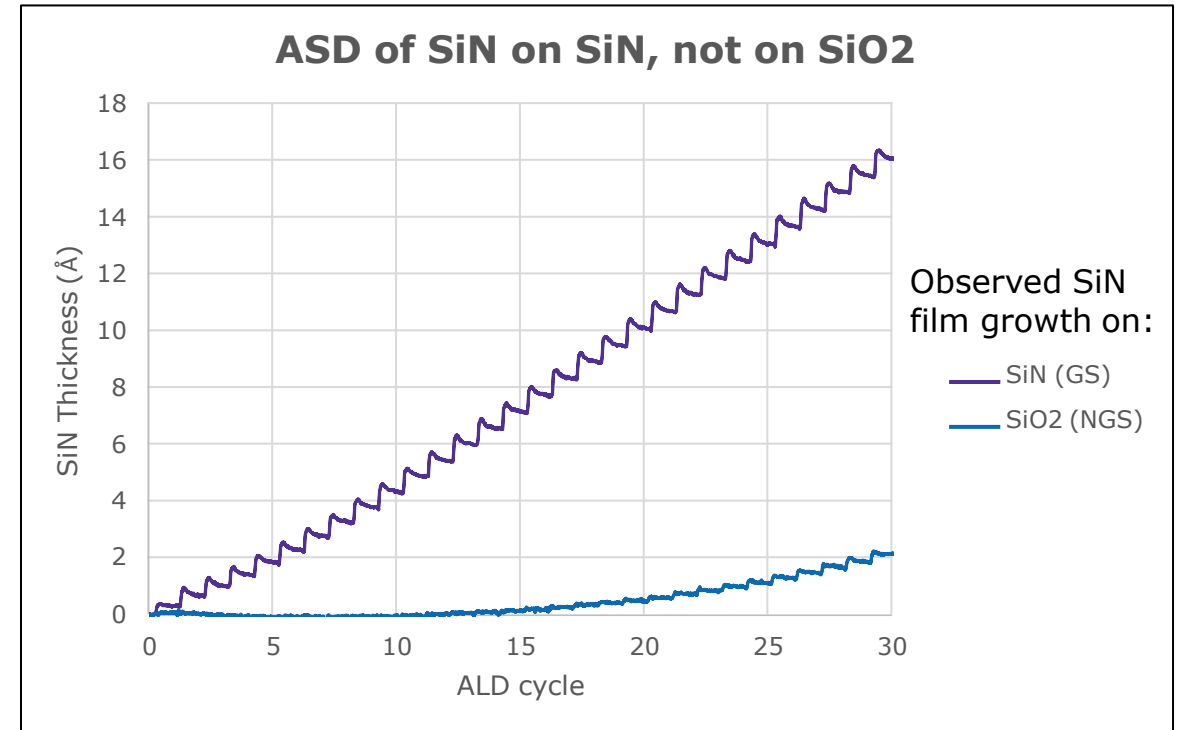
2. Passivate NGS selectively



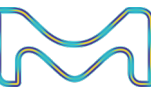
3. Deposit desired film on GS



4. Strip off inhibitor from NGS



$$* \text{ Selectivity} = \frac{Thk_{GS} - Thk_{NGS}}{Thk_{GS} + Thk_{NGS}} \rightarrow \text{Selectivity} = 78\% \text{ at } 16\text{\AA} \text{ for above example}$$

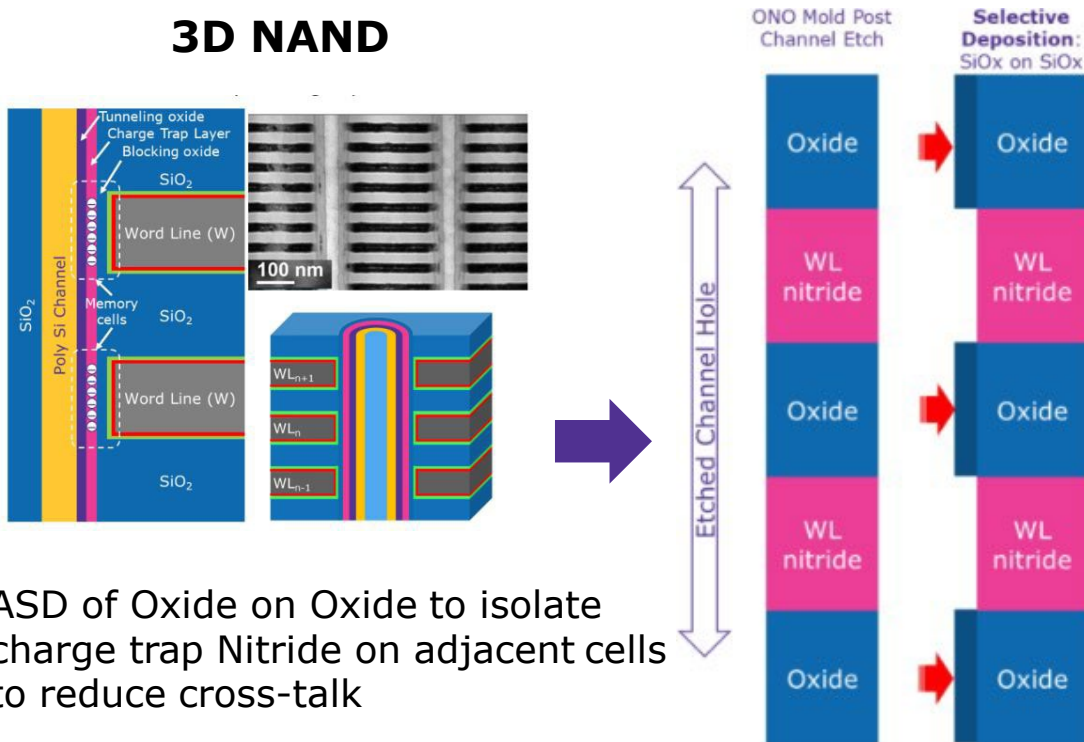


CHALLENGE 4

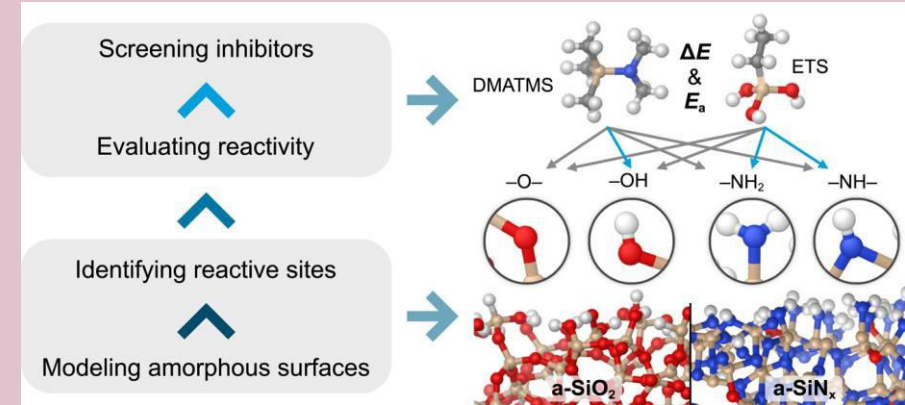
In-silico AI co-scientist for selective atomic layer growth schemes

Problem Statement: Passivate SiN (nitride) to deposit SiO_x on SiO_x (Oxide) with 90% selectivity at 10 nm of Oxide thickness

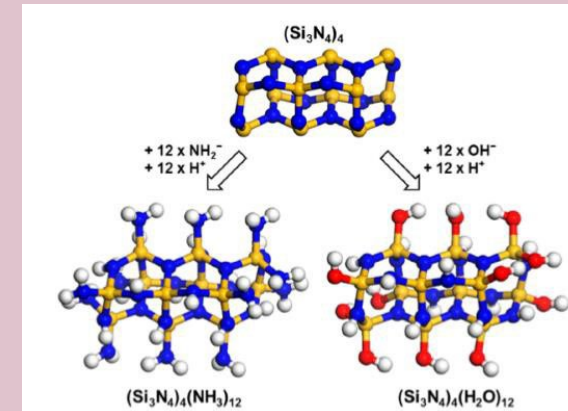
3D NAND



ASD of Oxide on Oxide to isolate charge trap Nitride on adjacent cells to reduce cross-talk



Applied Surface Science **2026**, 730, 166294



ACS Appl. Mater. Interfaces **2014**, 6, 10534–10541

Computational results are highly dependent on presumed structural model of amorphous surface!



CHALLENGE 4

In-silico AI co-scientist for selective atomic layer growth schemes

The in silico AI co-scientist shall include

1. An amorphous model surface builder that better reflects experimental surfaces

Note that models of amorphous

- SiO_x tend to have higher instances of reactive sites than what is realistic in experiments and
- SiN_x have $-\text{NH}_2$ and $-\text{NH}$ active sites with irregular spacing leading to excess calculations

Initial amorphous surfaces can be drawn from literature

2. An agentic selection logic for inhibitor/precursor candidates

Inhibitor/precursor chemistries can be drawn from literature

Layer chemistries (others welcome):

- non-/growth surface = silicon nitride, silicon oxide
- growing film = SiO_x , SiN_x or high-k oxides such as ZrO_2 , TiO_2 , HfO_2 , ...

Note: A markdown file may provide supplemental information and selection criteria for the agent

General Guideline:

Solutions should be programmed for ease of use with Python (Github packages, Python packages, Claude Code, etc. are all accepted)

